

E-Payment Systems (6 Hours)

1. Online Payment Cards:

- **Credit Cards:** Allow users to borrow money up to a certain limit to make purchases online.
- **Charge Cards:** Require users to pay the full balance at the end of each billing cycle.
- **Debit Cards:** Directly deduct funds from the user's bank account for payments.
- **Smart Cards:** Embedded with a microprocessor chip for secure transactions.

2. Processing Cards Online:

- Cardholder initiates payment.
- Payment gateway securely transmits data.
- Merchant's bank requests authorization.
- Issuer bank validates and approves transaction.

3. Credit Card Payment Procedure:

- **Authorization:** Verification of card details and availability of funds.
- **Authentication:** Secure verification through 3D Secure, OTP, etc.
- **Settlement:** Funds transferred from the issuing bank to the acquiring bank.

4. E-Micropayments:

- Small online transactions, typically under \$10.
- Used for digital goods (eBooks, songs, software downloads).
- Payment systems: PayPal, Apple Pay, Google Pay.

5. E-Checking and Its Processing Online:

- Digital version of traditional paper checks.
- Components: Routing number, account number, digital signature.
- Processed via Automated Clearing House (ACH) network.

6. Automated Clearing House (ACH) Network:

- Electronic payment system for batch processing.
- Used for direct deposits, bill payments, and business transactions.
- Provides cost-effective and secure transfers.

7. Mobile Payments (Digital Wallets):

- Payments made through mobile devices.

- Examples: Apple Pay, Google Pay, eSewa, Khalti.
- Mobile payment methods: QR code, NFC, SMS-based payments.

8. Mobile Payment Participants and Issues:

- Participants: Users, merchants, mobile payment service providers, financial institutions.
- Issues: Security risks, lack of infrastructure, regulatory concerns.

9. International Payments:

- Cross-border transactions.
- Methods: SWIFT, PayPal, Stripe.
- Challenges: Currency exchange rates, compliance with regulations.

10. Emerging EC Payment Systems and Issues:

- **Cryptocurrency:** Digital currency using blockchain technology (e.g., Bitcoin, Ethereum).
- **Virtual Currency:** In-game or platform-specific digital currencies.
- **Issues:** Volatility, regulatory challenges, security risks.

11. Case Study: Emerging Trends in Online Payment Systems in Nepal:

- Growth of digital wallets (eSewa, Khalti).
- Government initiatives promoting cashless transactions.
- Adoption of QR code payment systems.
- Challenges: Limited digital literacy, internet access issues.

This module explores the various aspects of electronic payment systems, highlighting their processes, technologies, and real-world applications, with a specific focus on Nepal's evolving digital payment landscape.